

# Vijay Raj Venugopal

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## PROFESSIONAL SUMMARY

Mechanical Engineer with hands-on experience in gas pipeline construction, mechanical systems installation, and field engineering. Proven track record applying ASME B31.3, BPVC, and industry codes to ensure quality and compliance on complex construction projects, with direct experience in pipeline installation, mechanical inspections, and coordinating field crews across all construction phases. Skilled in technical documentation, QC packages, and cross-functional team coordination. Holds active certifications in HVAC, Piping Design Engineering & Piping Isometrics, Six Sigma Green Belt, and FMEA, reflecting a commitment to engineering rigor and continuous improvement in construction and mechanical systems environments.

## EDUCATION

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|--------------------------------------------------|----------------------------|-----------|----------|
| Master of Engineering – Mechanical Engineering   | Old Dominion University    | GPA: 3.54 | Dec 2025 |
| Bachelor of Engineering – Mechanical Engineering | Bir Tikendrajit University | GPA: 3.33 | Dec 2023 |

## CERTIFICATIONS

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|--------------------------------------------------------------|----------------------------|----------|
| Piping Design Engineering & Piping Isometrics Masterclass    | Udemy                      | May 2026 |
| Six Sigma Green Belt (Lean Six Sigma – Accredited)           | Udemy (AIGPE)              | Apr 2026 |
| HVAC Maintenance from Zero to Hero (Dual HVAC Certification) | Udemy                      | Apr 2026 |
| AIGPE™ Certified FMEA Specialist                             | Udemy (AIGPE)              | Apr 2026 |
| Java Full Stack Development Certification                    | Quest Innovative Solutions | Apr 2023 |

## TECHNICAL SKILLS

Engineering & Design: AutoCAD, MATLAB, Gazebo, RViz2 | Codes & Standards: ASME B31.3, BPVC Section I Div VIII

Programming & Tools: Python, Java, Linux (Ubuntu) | MS Office (Excel, Word, PowerPoint)

Robotics & Sensing: ROS2, SLAM (Cartographer), Navigation2, LiDAR, Camera Systems, Ultrasonic Sensors

## PROFESSIONAL EXPERIENCE

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|------------------------------------------------------------------|---------------------|
| <b>Facilities Management Assistant – Old Dominion University</b> | Nov 2024 – Dec 2025 |
|------------------------------------------------------------------|---------------------|

- Managed 20+ facility work orders weekly across campus buildings, ensuring timely completion and minimal disruption.
- Coordinated with cross-functional maintenance teams to uphold safety standards and mechanical system reliability.
- Prioritized multiple concurrent service requests, improving task turnaround and overall operational workflow.

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|-----------------------------------------------|---------------------|
| <b>Engineer Trainee – STRIA Constructions</b> | Oct 2021 – Nov 2022 |
|-----------------------------------------------|---------------------|

- Monitored gas pipeline construction activities including trenching, joint coating, crossings, lowering, and backfilling; coordinated field equipment and personnel throughout all phases.
- Ensured compliance with engineering drawings, ASME and industry codes, and QC procedures; supported on-site mechanical inspections and formal documentation for client handover.
- Assisted in installation, alignment, and troubleshooting of mechanical components and pipeline materials using construction-grade field tools and equipment.
- Prepared progress reports and project documentation packages supporting quality control milestones and client acceptance.

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|----------------------------------------|-------------------------------------------|
| <b>Site Engineer – SAS Electricals</b> | Jun 2020 – Oct 2021   Sep 2023 – Dec 2023 |
|----------------------------------------|-------------------------------------------|

- Supervised installation of municipal water pipelines, ensuring strict adherence to project specifications and engineering drawings.
- Coordinated 10–20 field workers, managing daily site activities, task assignments, and scheduling.

- Conducted material inspections, trench depth measurements, and safety compliance checks throughout construction.

#### **Trainee Software Engineer – Quest Innovative Solutions**

Jan 2023 – Oct 2023

- Assisted in debugging and optimizing Java-based applications, improving system reliability.
- Participated in full-stack application development and collaborated with teams to test features and resolve issues.

## **PROJECTS**

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### **Autonomous Indoor Robot System (ROS2) – Master’s Project**

Built an autonomous robot using ROS2, SLAM (Cartographer), LiDAR, and Navigation2 for real-time indoor mapping and navigation. Integrated sensor fusion, path planning algorithms (A\*, Dijkstra), and real-time data processing pipelines.

### **Solar Assisted Heat Pump Water Heating System – Undergraduate Project**

Designed a hybrid system recovering waste heat from AC condensers combined with solar thermal collectors for efficient water heating; performed thermal analysis and system sizing.

## **LEADERSHIP & VOLUNTEERING**

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Y’s Men International – Youth Club President

2016 – 2022

Rise Up Forum – Volunteer

2019 – 2023

MRA – Volunteer

## **LANGUAGES**

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English (Professional) • Malayalam (Native) • Tamil (Conversational) • Hindi (Basic)